MINERAL RESERVE ESTIMATION USING SURPAC SOFTWARE

MI449 - MINE DESIGN PROJECT - II

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DECLARATION

We, hereby declare the report of the project work entitled "**MINERAL RESERVE ESTIMATION USING SURPAC SOFTWARE**" which is being submitted to the **National Institute of Technology Karnataka, Surathkal** for the award of the degree of **Bachelor of Technology** in Mining Engineering, is a bonafide report of the work carried out by us. The material contained in this report has not been submitted to any university or Institution for the award of any degree.

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CERTIFICATE

This is to certify that the B. Tech Project Work Report entitled "**MINERAL RESERVE ESTIMATION USING SURPAC SOFTWARE** " submitted by:

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as the record of the work carried out by them, is accepted as the B-Tech Project work report submission in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Mining Engineering of National Institute of Technology Karnataka, Surathkal.

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Table of Contents

Topic Name	Page No
1. Introduction	1
2. Literature Review	3
3. Methodology	6
3.1. Creation of a geological database	6
3.2. Display of boreholes	9
3.3. Styling of Drillholes	10
3.4. Sectioning and Digitising of Drill Holes	11
3.5. Ore Modelling	12
4. Conclusion	13
5. References	14

Mineral Reserve Estimation Using SURPAC Software

Chapter 1 - Introduction

The magnanimous growth and ever-increasing demand for finished products have pressurized the mother industry (mining) to supply raw materials to meet the increased demand for growth and sustainable development of society. These materials can be extracted from the earth's crust by any of the two methods of mining i.e., surface mining methods and underground mining methods. Surface mining operations can be classified into open pit, open cast, strip, alluvial, and in-situ mining, depending on the method of mining being considered (Hartman, 1987). Open-pit mining is a method of developing and exploiting the deposit by making a void in the surface, also known as a pit, and developing it sequentially to extract the ores. It starts off with a small pit and gradually develops with the lapse of time in a pre-planned manner to take the final shape of the pit called as ultimate pit which includes the initial void by which the deposit was attacked. Opencast mining is the advancement in the open pit mining methods, developed due to the acute shortage of granted leasehold areas where the overburden can be piled for future reclamation purposes. In the open-cast mining method, after all the ores in a particular area have been mined the overburden materials are dumped in-pit in those extracted areas instead of dumping it in the overburden pile within the leasehold area outside the pit. This has proved to be cost-effective and acquires lesser land area and continuous reclamation work is also ensured. At present opencast mining has a larger share of total ore production in India, owing to the technological advancement due to the availability of heavy earth moving machines (HEMMs) which has increased the rate of production while ensuring safety and requires lesser manpower. The increased rate of operation facilitates lesser development time.

The advancement in computer technology has started the development of software applications. SURPAC is one of the software applications that was developed to be used in the mining industry. This SURPAC software should be addressed by a specialized staff composed of mining engineers, surveyors, and geologists. Each of them has a separate module dedicated. The surveyors create a Digital Terrain Model (DTM), geologists create a digital model of the deposit, and mining engineers use it to estimate the production of useful minerals. Underground minerals, metals, and nonmetals are invisible whose shapes, quality compositions, and quantities are not known. Geological explorations and investigations aimed at determining the unknown minerals and metals.

At the start of the process, topographical and lithology data are gathered and a database is generated by using SURPAC. Depth, thickness, and grade changes, ore volume, shape and extensions, and properties are determined by mathematical and algorithm approaches using this database. All numerical estimations are used to get out visuals to bring out the body model. The concrete data to define the shape, location, quality, and quantity of an ore body is by bore cores. GPS data is usually taken to draw topographical maps and surfaces. Underground maps like thickness and grade contours are drawn also. Topographical coordinates are combined with stratigraphic information, a 3-dimensional data set is handled. After following several mathematical techniques, 3- a dimensional ore body model is often obtained. Besides the physical ore model, the quality composition should even be known

This is often crucial because further engineering activities have a cheap aspect to perform. With the known volume of block properties like thickness and grade of ore at each particular block, it becomes possible to convert this information to an economical aspect. (Volume × tonnage factor × grade = block reserve.) Surveying includes 3-dimensional components x, y, and z (easting, northing, elevation) which are used for surface modeling. Drill hole data depth and layer information contribute to explaining how the geological structure is in the three dimensions. Drill holes also carry knowledge about ore grade. Geological interpretation of stratigraphic layers provides a 3-dimensional ore body model.

Objective

- To Study the Mineral Resource Estimation of a particular mining project using SURPAC software.
- Develop a model in SURPAC software using Geology, Survey, Borehole data, and Assay Value of a particular deposit.

Chapter 2 - Literature Review

Mineral reserve estimation is a critical aspect of the mining industry, with accurate estimates essential for effective decision-making and resource management. Surpac software is widely used in the industry for its advanced capabilities in geological modeling and reserve estimation. Previous studies have highlighted the benefits of using Surpac software, such as its user-friendly interface, efficient data handling, and robust algorithms for resource estimation. Case studies have demonstrated the software's effectiveness in producing reliable reserve estimates, leading to improved mine planning and operational efficiency.

Comparative studies have also been conducted to evaluate Surpac against other conventional methods and mathematical models, showing its competitive advantages in certain aspects of mineral reserve estimation. However, challenges such as data integration and model validation have been identified, suggesting areas for further research and development. Future trends in mineral reserve estimation using Surpac software point towards advancements in machine learning, automation, and integration with other technologies for more accurate and efficient resource assessment. Overall, the literature review underscores the importance of Surpac software in mineral reserve estimation and highlights opportunities for enhancing its capabilities in the field.

2.2 COMPUTER AIDED ORE RESERVE ESTIMATION

With the progression of technology and the advent of computers, software developers applied their expertise to link the fundamentals of the estimation process with mathematical functions and equations. This effort resulted in the creation of sophisticated software designed for estimating ore reserves with increased speed and enhanced accuracy. Interestingly, these software applications still incorporate the three fundamental principles of conventional methods outlined by Popoff (1966).

Additionally, geostatistical tools emerged as valuable resources for ore reserve estimation through computer applications. The three commonly employed techniques in computer-assisted ore reserve estimation include nearest neighbor, inverse distance, and kriging.

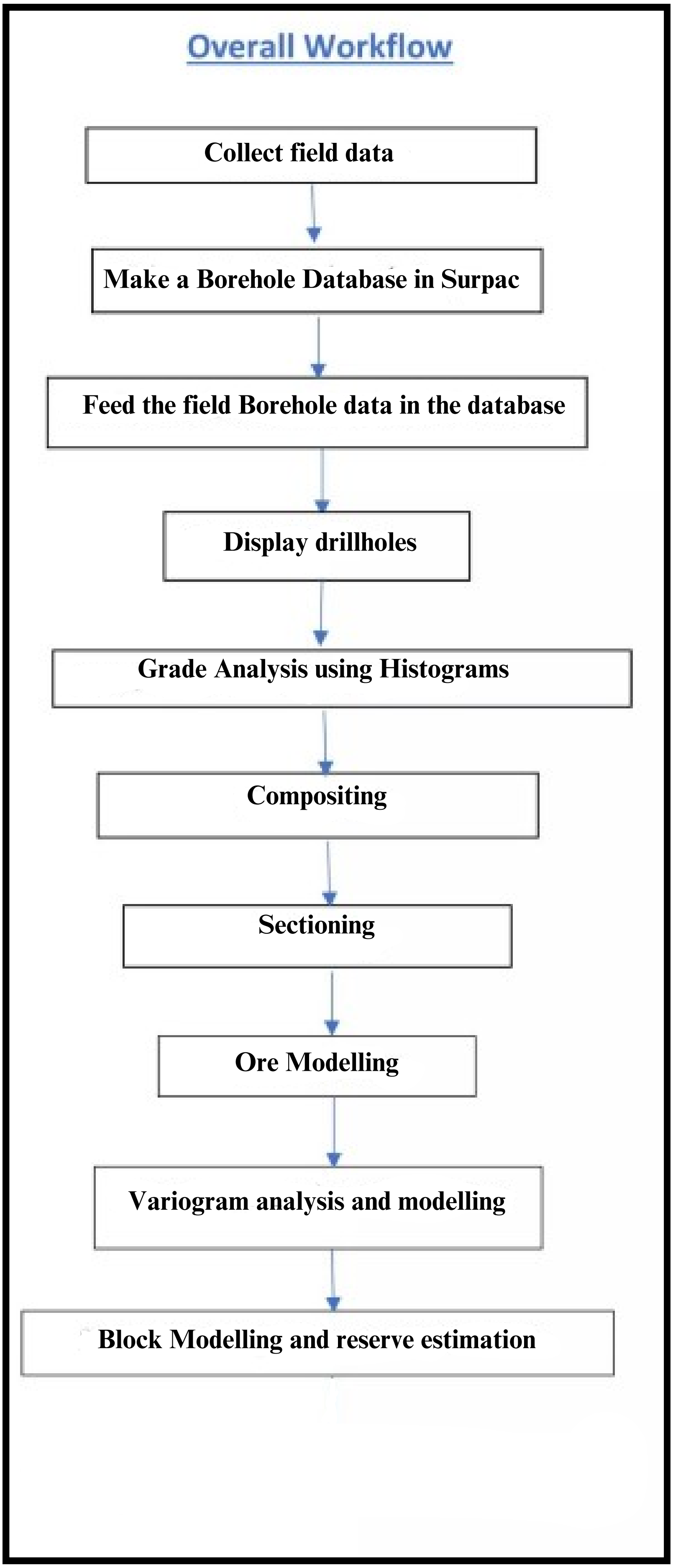
2.2.1 Surpac:

In this project, we have used GEOVIA Surpac for studying and further planning of the mine using the data provided by the instructor.

GEOVIA Surpac is the world's most popular geology and mine planning software, supporting open pit and underground operations and exploration projects.

Using Surpac, we can visualize an interpolation, lock it in 3-D, and then we can bring in all the drilling, all the infrastructure, the development, the headings, and know instantly whether we are on track.

It also allows us to integrate various processes to result in a well-designed plan.


Overall Workflow
Collect field data
Make a Borehole Database in Surpac
Feed the field Borehole data in the database
Display drillholes
Grade Analysis using Histograms
Compositing
Sectioning
Ore Modelling
Variogram analysis and modelling
Block Modelling and reserve estimation

Chapter 3 - Methodology

3.1. CREATION OF A GEOLOGICAL DATABASE

For the process of geological modeling, the first step is to create a geological database. The geological database is basically done from the raw data acquired from the boreholes or equivalently drill holes so that the software understands the type of information fed into its system and can compute the results or display any graphical form according to the user's requirements. For feeding into a Surpac database or basically any geological database two mandatory tables are required - collar and survey. For obtaining information related to the grade, types of rocks present near the orebody, and the presence of any geological features or discontinuities, other optional tables are used such as assay, litho, geology, etc. The software system uses two tables intrinsically as a part of the system tables - translation and styles which deal with the graphical display formats.

During the process of data importing, data is taken from the CSV files (the given data files) to the newly created Surpac database through the software wizard. Data can be imported from access and various versions of Paradox database files. It has to be made sure that the correct CSV file should be selected for the tables and the column number of Surpac database parameters and the field number of the parameters mentioned in CSV files should match each other.

The starting point of all mining projects is the drill-hole data. It continues the basis on which feasibility studies and ore reserve estimation are done. A number of tables are included in a geological database, containing different data. Each table contains a number of fields, having many records, and with each record containing the data fields. Surpac uses a relational database model and supports several different types of databases, i.e. oracle, paradox, and Microsoft Access.

Two mandatory tables are required in Surpac within a database:

1. Collar
2. Survey.

3.1.1 Collar table:

The information in this table describes the location of the drill-hole collar, the maximum depth of the hole, and whether to calculate a linear or curved hole trace when retrieving the hole. For each drill hole, optional collar data may also be stored. For instance, data drilled, type or drill-hole, or project name.

In a collar table, the mandatory fields are as given below:

1. Hole_id
2. Y
3. X
4. Z
5. Max_depth
6. Hole_path

3.1.2 Survey table:

It stores the drill-hole survey information, used for calculating the drill-hole terrace co-ordinates.

The mandatory field in a survey table includes:

1. Down-hole survey depth,
2. Dip, and
3. Azimuth of the hole.

Other Optional Tables:

Apart from the mandatory tables, the optional tables include **Geology** and **Assay**. These tables are added and used to store information.

3.1.3 Assay Table: Assay table is an Interval optional table (depth from to depth to) and assay table store information about the concentration of different components in an interval. Mandatory fields for Assay table: hole_id, depth_from, depth_to Optional fields that we have for assay table: percentages of fe, sio2, Al2Os3, loi, p in an interval.

3.1.4 Geology Table: It is also an Interval optional table. And it stores information about different geology in an interval. And the different geology of the field store as a code (0-10) known as lithocode(Icode). Mandatory fields for the geology table: hole_id, depth_from, depth_to, Icode

3.1.2 Importing Data

While importing data two things are important to pay attention to, such as

1)Naming the format file

2)Checking the "overlapping sample check"

The name of the format file should be the same as the database name for simplicity as both files are required for use in other systems.

Overlapping of data may occur between the assay and lithology data as they refer to the same interval of a borehole. The process of lithology and assay sampling is elaborated in Fig. 2. Therefore checking of the overlapping should be done.

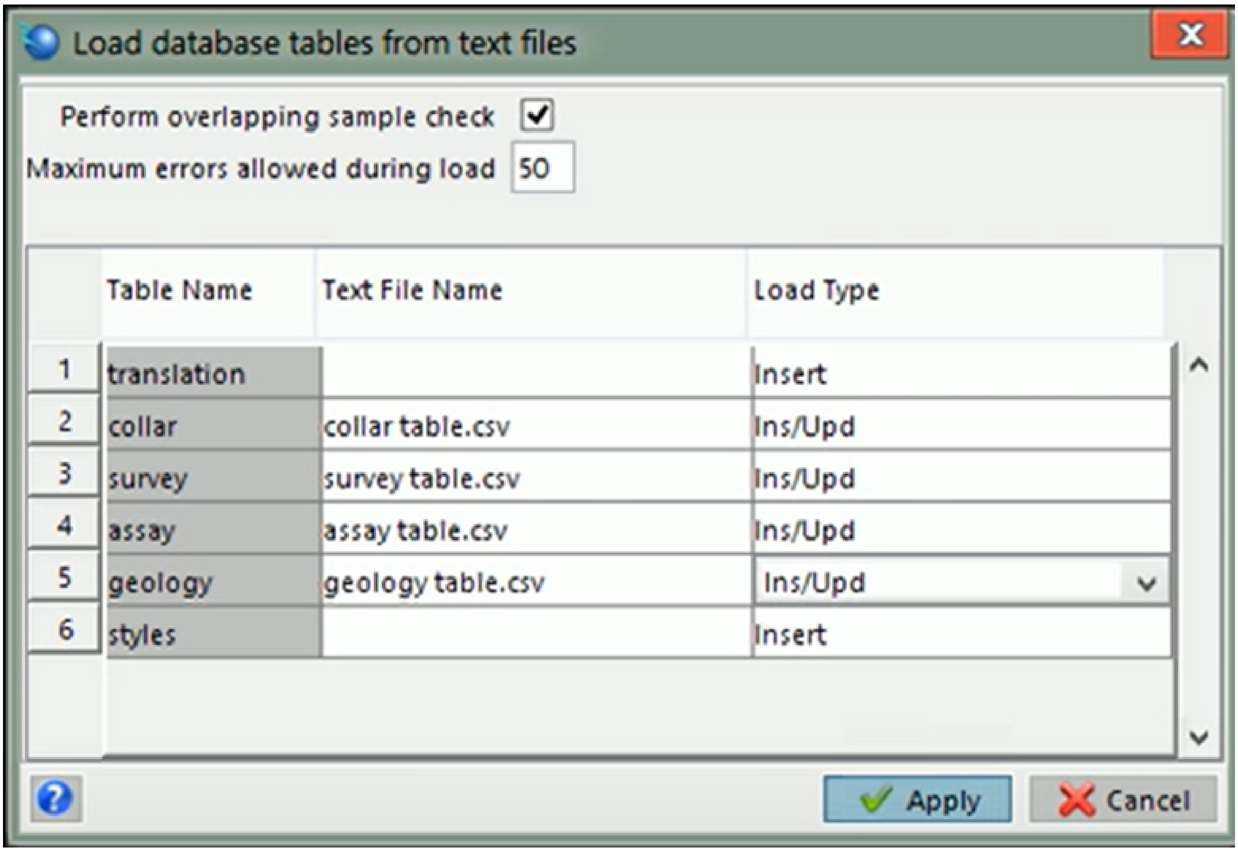


Fig 1 .Importing the data from CSV files

3.2. DISPLAY OF BOREHOLES

The mine plan was imported into the Surpac graphics in the form of string file, i.e. str format. Then the mine plan was corrected for its scale and orientation by using the 'Move string constrained by – transformation' function. The map was rotated to an angle of +72° and its scale was increased to a magnitude of 13.6. Simultaneously the map was also subjected to translation till it reached its desired coordinates.

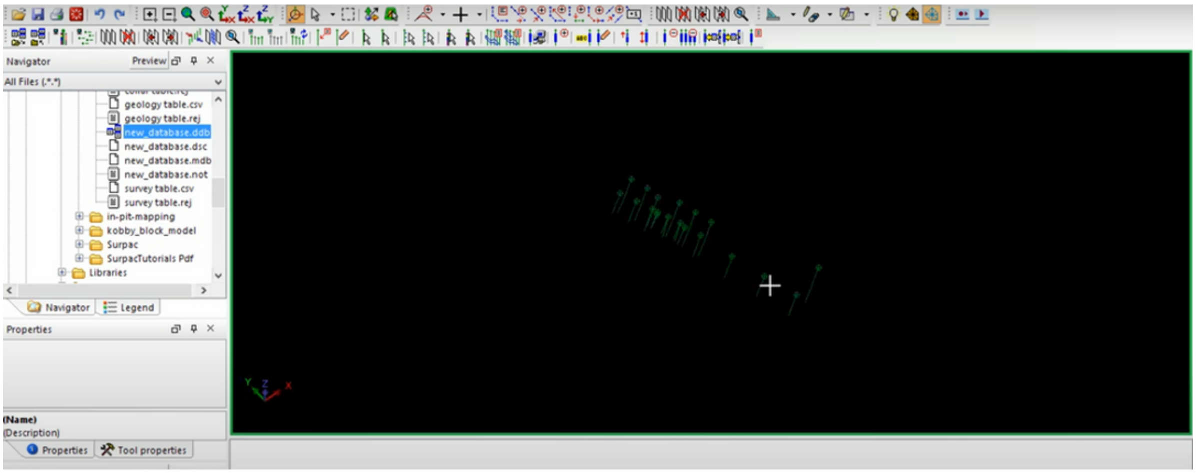


Fig 2 .Display of Boreholes

A geological database consists of a number of tables, each of which contains different types of data. Each table contains a number of fields. Each table also has many records, with each record containing the data fields. Surpac uses a relational database model and supports several types of databases, including Oracle, Paradox, and Microsoft Access. Surpac also supports Open Database Connectivity (ODBC) and can connect to databases across networks. A database can contain up to 50 tables and each table can have a maximum of 60 fields. Surpac requires two mandatory tables within a database: collar and survey. Besides the two mandatory tables, two optional tables namely: assay and geology may be required. The data available in this table are either saved in the "Microsoft Access" file or in the Comma Separated Value (CSV) files of Microsoft Excel. In addition to these tables, surpac also uses the translation and the styles table by default. Entering the above-mentioned individual data which were prepared in the Comma Separated Value (CSV) files of Microsoft Excel to the surpac software, a geological database was created which contains 71 drill holes as shown below.

3.3 Styling Drill Holes:

For the batter visualization and differentiate between the grades.

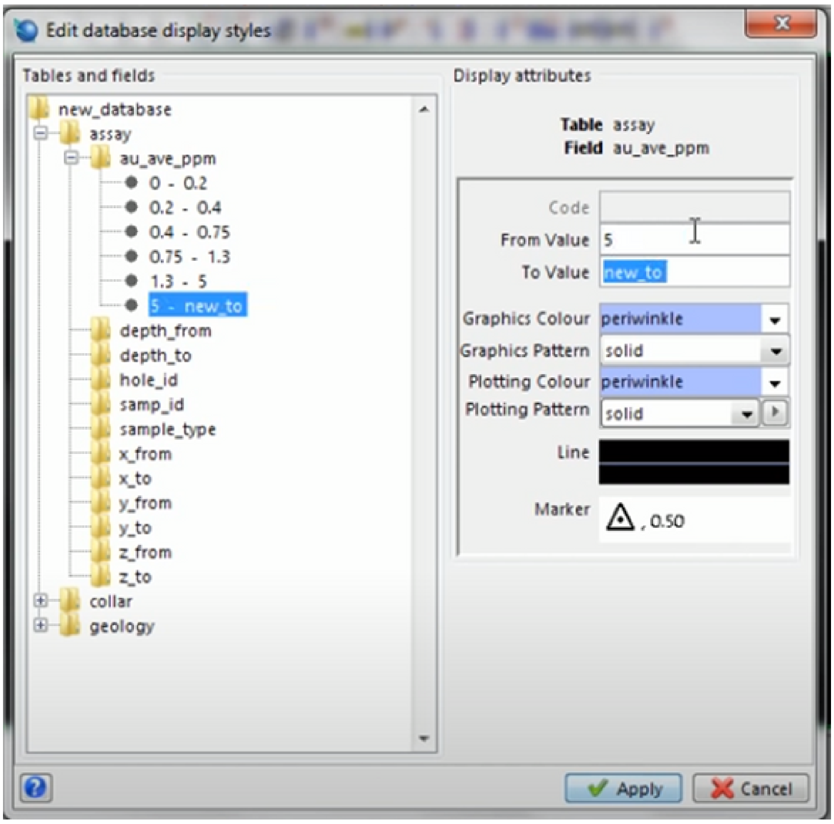


Fig 3. Applying styles to the drill holes

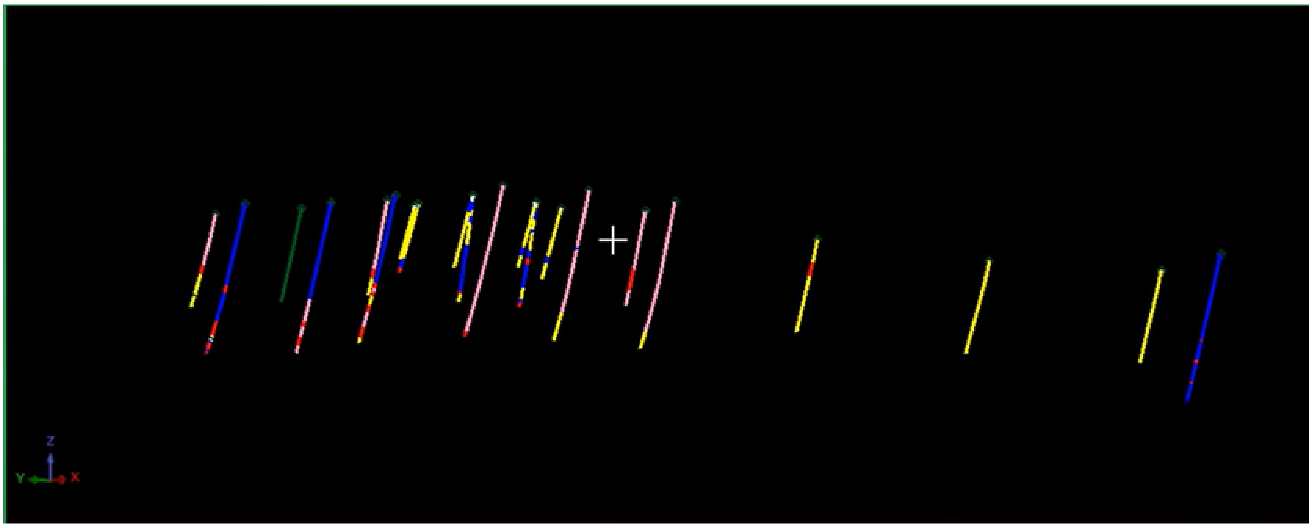


Fig 4 Styling Drill Holes.

3.4 Sectioning and Digitising of Drill Holes:

Terminologies: -

Sectioning: Sectioning is the process of creating a mirror image of the drill holes when cut through by a plane.

Digitizing: digitising is the act of outlining the ore terrain or mineralization. We will use this to define the extent of the ore body based on the direction and length of drill holes.

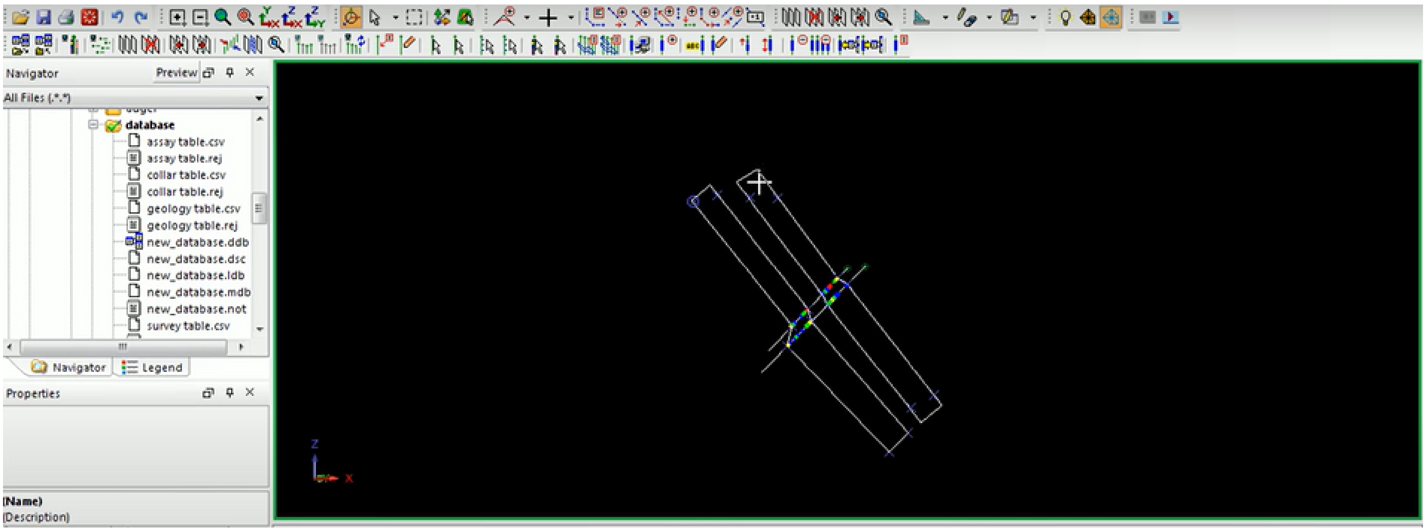


Fig 5.Sectioning and Digitising of Drill Holes

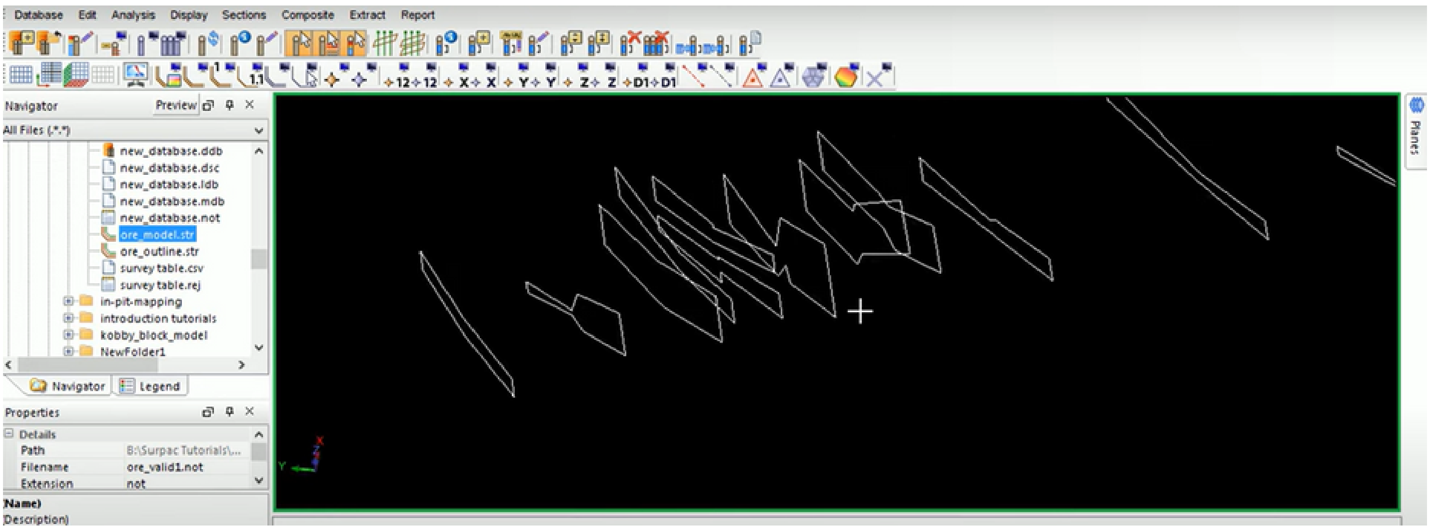


Fig 6. segments after digitising

3.5. ORE MODELLING

Prior to 3-D modeling, some of the other functions to be carried out include the generation of surface DTM from a given borehole set, indicating the depths in the boreholes which can be considered as ore, and the creation of sections for different drill hole sets. The surface string is generated using the 'import data from one file' function.

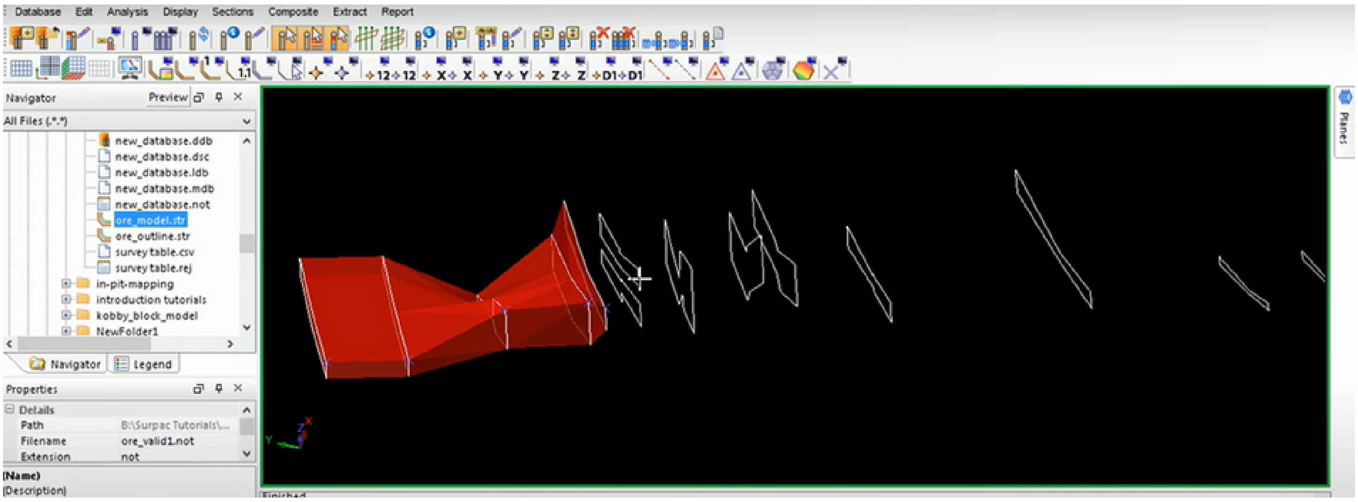


Fig 7.1. 3-D ore body model

A 3D model of the ore body is the best resource for visualizing the characteristics of the deposit. The ore body was developed by triangulating the segments of the mineralization zone which were sectioned and digitized from the boreholes. The segments that are to be used for developing the 3D model of the orebody should be first cleaned. The cleaning process should be fulfilled on the string of the segments. Out of all the tests, 3 tests (i.e. cross-overs, duplicated points, and spikes) could fulfill the requirements. After completing of the tests, the segments are joined together as per the requirements and behavior of the segments. Then, the newly formed 3D model is validated, if the validation is successful the volume and the area of the orebody can then be obtained by producing the "report volume of solids".

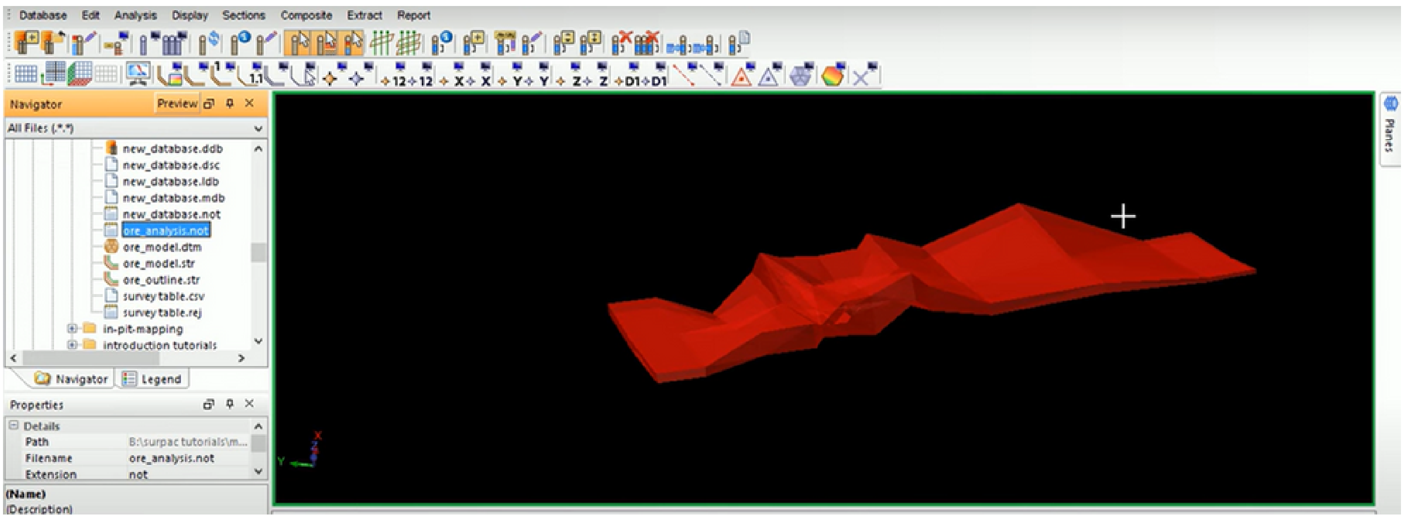


Fig 7.2. 3-D ore body model

4. Conclusion

The mineral reserves estimation project utilizing SURPAC software has proven to be an invaluable asset in the mining industry, significantly enhancing the accuracy, efficiency, and reliability of reserves assessments. The importance of reserves estimation cannot be overstated, as it plays a pivotal role in various stages of mining operations, from initial exploration to mine planning and resource optimization.

Reserves estimation serves as the foundation for informed decision-making throughout the mining life cycle. During exploration, it provides a basis for assessing the economic viability of a potential site, guiding investment decisions and risk management.

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